GEMM Analytical Model - Equations

Auto-generated from PYTHON_FORMULAS and LATEX_SYMBOL_OVERRIDES in gemmanalytics.py.

Output order: execution

Machine Parameters (Pre-Defined)

$f_{GHz}^{(GT)}$: [row 8] GT Freq (GHz)

$|XeCU|$: [row 26] XeCU count

$|XeCore|_{perXeCU}$: [row 27] XeCore per XeCU

$|EU|_{perXeCore}$: [row 28] EU per XeCore

$|L2Banks|_{perXeCU}$: [row 29] L2 banks per XeCU

$L2BankSize_{MB}$: [row 30] bank capacity (MB)

D_{DPAS} : [row 32] DPAS depth

$\eta_{systolic}$: [row 35] compute efficiency %

$L1RdBW_{Bp(Clk\cdot EU)}^{max}$: [row 86] L1 read max (B/EU/clk)

$L1WrBW_{Bp(Clk\cdot EU)}^{max}$: [row 87] L1 write max (B/EU/clk)

$GtiRdBW_{BpClk}^{max}$: [row 97] GTI read max BW (B/clk)

$GtiWrBW_{BpClk}^{max}$: [row 98] GTI write max BW (B/clk)

$MemBW_{GBps}^{max}$: [row 124] max possible HBM BW (GB/s)

$DPAS_{MixFmt}$: [row 1000] native DPAS mixed-precision support

Workload Parameters (Pre-Defined)

$Fmt^{(matA)}$: [row 3] input A data format (e.g. fp8)

$Fmt^{(matB)}$: [row 4] input B data format (e.g. fp4)

$Fmt^{(matD\downarrow)}$: [row 5] output D data format (e.g. fp8)

M_{dim} : [row 9] M dimension

K_{dim} : [row 10] K dimension

N_{dim} : [row 11] N dimension

$Byte_{perElement}^{(matD)}$: [row 24] output bytes per element (fp32)

ρ_{GT} : [row 34] machine occupancy %

$M_{perThread}$: [row 39] M per thread

$K_{perThread}$: [row 40] K per thread

$N_{perThread}$: [row 41] N per thread

$W_{thread}^{(ThreadGroup)}$: [row 45] TG width in units of thread

$H_{thread}^{(ThreadGroup)}$: [row 46] TG height in units of thread

$W_{TG, keep\ cluster\ size\ as\ 4}^{(XeCoreCluster)}$: [row 50] XeCore cluster width in units of TG (keep cluster size as 4)

$H_{TG, keep\ cluster\ size\ as\ 4}^{(XeCoreCluster)}$: [row 51] XeCore cluster height in units of TG (keep cluster size as 4)

$W_{TG}^{(XECUTile)}$: [row 54] XeCU tile width in units of TG

$H_{TG}^{(XECUTile)}$: [row 55] XeCU tile height in units of TG

$W_{XeCU}^{(GPUTile)}$: [row 58] GPU tile width in XeCU unit

Computed Equations

row 22 - INPUT A BYTES PER ELEMENT

$$\mathrm{Bytes}_{perElement}^{(matA)} = \mathrm{DataFormatToBytes}[\mathrm{Fmt}^{(matA)}]$$

row 23 - INPUT B BYTES PER ELEMENT

$$\mathrm{Bytes}_{perElement}^{(matB)} = \mathrm{DataFormatToBytes}[\mathrm{Fmt}^{(matB)}]$$

row 25 - OUTPUT BYTES PER ELEMENT AFTER DOWN CONVERSION

$$\mathrm{Bytes}_{perElement}^{(matD\downarrow)} = \mathrm{DataFormatToBytes}[\mathrm{Fmt}^{(matD\downarrow)}]$$

row 33 - EU COUNT

$$|\mathrm{EU}| = |\mathrm{XeCore}|_{perXeCU} \times |\mathrm{EU}|_{perXeCore} \times |\mathrm{XeCU}|$$

row 4002 - MIXED PRECISION DPAS THROUGHPUT DELTA

$$\delta_{\tau_{DPAS}} = \mathrm{DPAS}_{MixFmt} \wedge \left(\mathrm{Fmt}^{(matA)} \in \{\mathrm{i8}\} \wedge \mathrm{Fmt}^{(matB)} \in \{\mathrm{i2}, \mathrm{i4}\} \vee \mathrm{Fmt}^{(matA)} \in \{\mathrm{i2}, \mathrm{i4}\} \wedge \mathrm{Fmt}^{(matB)} \in \{\mathrm{i8}\}\right)$$

row 4001 - MMA MAC THROUGHPUT PER CHANNEL STAGE

$$\tau_{mMACper(Clk \cdot CHANNEL \cdot STAGE)}^{(peak)} = \frac{4}{max(\mathrm{Bytes}_{perElement}^{(matA)}, \mathrm{Bytes}_{perElement}^{(matB)})} \times \left(1 + \delta_{\tau_{DPAS}}\right)$$

row 4000 - MMA MAC THROUGHPUT PER EU

$$\tau_{mMACper(Clk \cdot EU)}^{(peak)} = \tau_{mMACper(Clk \cdot CHANNEL \cdot STAGE)}^{(peak)} \times \mathrm{D}_{DPAS} \times 16$$

row 36 - MMA MAC THROUGHPUT PER XECORE

$$\tau_{mMACper(Clk \cdot XeCore)}^{(peak)} = \tau_{mMACper(Clk \cdot EU)}^{(peak)} \times |\mathrm{EU}|_{perXeCore}$$

row 42 - CLKS PER DPAS

$$\mathrm{CLKS}_{DPAS} = \frac{\mathrm{M}_{perThread} \times \mathrm{K}_{perThread} \times \mathrm{N}_{perThread}}{\frac{\tau^{(peak)}_{mMACper(Clk\cdot XeCore)}}{|\mathrm{EU}|_{perXeCore}}}$$

row 43 - THREAD WIDTH IN UNITS OF ELEMENTS

$$\mathrm{W}^{(Thread)}_{element} = \mathrm{N}_{perThread}$$

row 44 - THREAD HEIGHT IN UNITS OF ELEMENTS

$$\mathrm{H}^{(Thread)}_{element} = \mathrm{M}_{perThread}$$

row 47 - TG WIDTH IN UNITS OF ELEMENT REALIZED BY MULTIPLE MMA ITERATIONS

$$\mathrm{W}^{(ThreadGroup, realized\,by\,multiple\,MMA\,iterations)}_{element} = \mathrm{W}^{(ThreadGroup)}_{thread} \times \mathrm{W}^{(Thread)}_{element}$$

row 14 - TG TILES IN N

$$|\mathrm{Tiles}|^{(TG)}_{N} = \frac{\mathrm{N}_{dim}}{\mathrm{W}^{(ThreadGroup, realized\,by\,multiple\,MMA\,iterations)}_{element}}$$

row 48 - TG HEIGHT IN UNITS OF ELEMENT

$$\mathrm{H}^{(ThreadGroup)}_{element} = \mathrm{H}^{(ThreadGroup)}_{thread} \times \mathrm{H}^{(Thread)}_{element}$$

row 15 - TG TILES IN M

$$|\mathrm{Tiles}|^{(TG)}_{M} = \frac{\mathrm{M}_{dim}}{\mathrm{H}^{(ThreadGroup)}_{element}}$$

row 52 - XECORE CLUSTER WIDTH IN UNITS OF ELEMENT

$$\mathrm{W}^{(XeCoreCluster)}_{element} = \mathrm{W}^{(XeCoreCluster)}_{TG, keep\,cluster\,size\,as\,4} \times \mathrm{W}^{(ThreadGroup, realized\,by\,multiple\,MMA\,iterations)}_{element}$$

row 16 - TG CLUSTER TILES IN N

$$|\mathrm{Tiles}|^{(TG\,Cluster)}_{N} = \frac{\mathrm{N}_{dim}}{\mathrm{W}^{(XeCoreCluster)}_{element}}$$

row 53 - XECORE CLUSTER HEIGHT IN UNITS OF ELEMENT

$$\mathrm{H}^{(XeCoreCluster)}_{element} = \mathrm{H}^{(XeCoreCluster)}_{TG, keep\,cluster\,size\,as\,4} \times \mathrm{H}^{(ThreadGroup)}_{element}$$

row 17 - TG CLUSTER TILES IN M

$$|\mathrm{Tiles}|^{(TG\,Cluster)}_{M} = \frac{\mathrm{M}_{dim}}{\mathrm{H}^{(XeCoreCluster)}_{element}}$$

row 56 - XECU TILE WIDTH IN UNITS OF ELEMENT

$$\mathrm{W}_{element}^{(XeCUTile)} = \mathrm{W}_{TG}^{(XECUTile)} \times \mathrm{W}_{element}^{(ThreadGroup, realized\, by\, multiple\, MMA\, iterations)}$$

row 18 - XECU TILES IN N

$$|\mathrm{Tiles}|_{N}^{(XeCU)} = \frac{\mathrm{N}_{dim}}{\mathrm{W}_{element}^{(XeCUTile)}}$$

row 20 - GPU TILES IN N

$$|\mathrm{Tiles}|_{N}^{(GPU)} = \frac{\frac{\mathrm{N}_{dim}}{\mathrm{W}_{element}^{(XeCUTile)}}}{\mathrm{W}_{XeCU}^{(GPUTile)}}$$

row 57 - XECU TILE HEIGHT IN UNITS OF ELEMENT

$$\mathrm{H}_{element}^{(XeCUTile)} = \mathrm{H}_{TG}^{(XECUTile)} \times \mathrm{H}_{element}^{(ThreadGroup)}$$

row 19 - XECU TILES IN M

$$|\mathrm{Tiles}|_{M}^{(XeCU)} = \frac{\mathrm{M}_{dim}}{\mathrm{H}_{element}^{(XeCUTile)}}$$

row 59 - GPU TILE HEIGHT IN XECU UINT

$$\mathrm{H}_{XeCU}^{(GPUTile)} = \frac{|\mathrm{XeCU}|}{\mathrm{W}_{XeCU}^{(GPUTile)}}$$

row 21 - GPU TILES IN M

$$|\mathrm{Tiles}|_{M}^{(GPU)} = \frac{\frac{\mathrm{M}_{dim}}{\mathrm{H}_{element}^{(XeCUTile)}}}{\mathrm{H}_{XeCU}^{(GPUTile)}}$$

row 13 - WAVES

$$\mathrm{N}_{waves} = \mathrm{CEILING}(|\mathrm{Tiles}|_{N}^{(GPU)}, 1) \times \mathrm{CEILING}(|\mathrm{Tiles}|_{M}^{(GPU)}, 1)$$

row 60 - GPU TILE WIDTH IN UNITS OF ELEMETNS

$$\mathrm{W}_{element}^{(GPUTile)} = \mathrm{W}_{XeCU}^{(GPUTile)} \times \mathrm{W}_{element}^{(XeCUTile)}$$

row 61 - GPU TILE HEIGHT IN UNITS OF ELEMETNS

$$\mathrm{H}_{element}^{(GPUTile)} = \mathrm{H}_{XeCU}^{(GPUTile)} \times \mathrm{H}_{element}^{(XeCUTile)}$$

row 63 - MAT A INPUT SIZE B

$$\mathrm{Size}_{B}^{(matA)} = \mathrm{M}_{dim} \times \mathrm{K}_{dim} \times \mathrm{Bytes}_{perElement}^{(matA)}$$

row 64 - MAT B INPUT SIZE B

$$\text{Size}_B^{(matB)} = \text{K}_{dim} \times \text{N}_{dim} \times \text{Bytes}_{perElement}^{(matB)}$$

row 65 - MAT C INPUT D OUTPUT SIZE B

$$\text{Size}_B^{(matC,matD)} = \text{M}_{dim} \times \text{N}_{dim} \times \text{Bytes}_{perElement}^{(matD\downarrow)}$$

row 66 - MAT D INTERMEDIATE SIZE B

$$\text{Size}_B^{(matD\downarrow)} = \text{M}_{dim} \times \text{N}_{dim} \times \text{Byte}_{perElement}^{(matD)}$$

row 102 - TOTAL L2 SIZE B FOR A SINGLE INSTANCE

$$\text{L2Size}_B = \text{L2BankSize}_{MB} \times |\text{L2Banks}|_{perXeCU} \times 1024 \times 1024$$

row 103 - WORKING DATA SET SIZE OF K IN L2 CORRESP 20K CLOCKS OF THREAD DIVERGENCE

$$|\text{WorkingSet}|_{20K\ clks\ of\ thread\ divergence}^{(K\ in\ L2)} = \min(20000 \times \frac{\text{K}_{perThread}}{\text{CLKS}_{DPAS}}, \text{K}_{dim})$$

row 104 - TOTAL REQUIRED L2 SIZE FOR IDEAL HIT RATE B FOR A SINGLE INSTANCE AND SINGLE WAVE

$$\begin{aligned}\text{TotalRequiredL2Size}_{matB\ alwayshit}^{1\ instance,\ 1\ wave} = {} & \text{H}_{element}^{(XeCUTile)} \times \text{Bytes}_{perElement}^{(matA)} \times |\text{WorkingSet}|_{20K\ clks\ of\ thread\ divergence}^{(K\ in\ L2)} \\ & + \text{W}_{element}^{(XeCUTile)} \times \text{Bytes}_{perElement}^{(matB)} \times |\text{WorkingSet}|_{20K\ clks\ of\ thread\ divergence}^{(K\ in\ L2)} \\ & + \text{W}_{element}^{(XeCUTile)} \times \text{H}_{element}^{(XeCUTile)} \times \text{Bytes}_{perElement}^{(matD\downarrow)}\end{aligned}$$

row 125 - MAX POSSIBLE HBM BW FREQ B CLK

$$\text{MemBW}_{BpClk}^{max} = \frac{\text{MemBW}_{GBps}^{max}}{\text{f}_{GHz}^{(GT)}}$$

row 5000 - WORKLOAD MAC PER XECORE

$$\text{WL}_{MAC}^{(XeCore)} = \frac{\text{M}_{dim} \times \text{K}_{dim} \times \text{N}_{dim}}{|\text{XeCore}|_{perXeCU} \times |\text{XeCU}|}$$

row 37 - CLK SPECIFIED EFFICIENCY

$$\text{T}_{clk}^{(total)} = \frac{\text{WL}_{MAC}^{(XeCore)}}{\tau_{mMACper(Clk \cdot XeCore)}^{(peak)} \times \eta_{systolic}}$$

row 69 - TOTAL L2 READ B

$$\mathrm{L2Rd}_B^{(total)} = \mathrm{Size}_B^{(matA)} \times \mathrm{CEILING}(\frac{\mathrm{N}_{dim}}{\mathrm{W}_{element}^{(ThreadGroup, realized\,by\,multiple\,MMA\,iterations)}}, 1)$$
$$+\mathrm{Size}_B^{(matB)} \times \mathrm{CEILING}(\frac{\mathrm{M}_{dim}}{\mathrm{H}_{element}^{(ThreadGroup)}}, 1)$$

row 70 - TOTAL L2 WRITE B

$$\mathrm{L2Wr}_B^{(total)} = \mathrm{Size}_B^{(matC, matD)}$$

row 71 - TOTAL L1 READ B

$$\mathrm{L1Rd}_B^{(total)} = \mathrm{Size}_B^{(matA)} \times \mathrm{CEILING}(\frac{\mathrm{N}_{dim}}{\mathrm{W}_{element}^{(Thread)}}, 1)$$
$$+\mathrm{Size}_B^{(matB)} \times \mathrm{CEILING}(\frac{\mathrm{M}_{dim}}{\mathrm{H}_{element}^{(Thread)}}, 1)$$

row 72 - TOTAL L1 WRITE B

$$\mathrm{L1Wr}_B^{(total)} = \mathrm{Size}_B^{(matC, matD)}$$

row 74 - L2 READ B XECORE CLK

$$\mathrm{L2RdBW}_{Bp(Clk \cdot XeCore)} = \frac{\frac{\mathrm{L2Rd}_B^{(total)}}{|\mathrm{XeCU}| \times |\mathrm{XeCore}|_{perXeCU}}}{\mathrm{T}_{clk}^{(total)}}$$

row 75 - L2 WRITE B XECORE CLK

$$\mathrm{L2WrBW}_{Bp(Clk \cdot XeCore)} = \frac{\frac{\mathrm{L2Wr}_B^{(total)}}{|\mathrm{XeCU}| \times |\mathrm{XeCore}|_{perXeCU}}}{\mathrm{T}_{clk}^{(total)}}$$

row 76 - L2 READ WRITE B XECORE CLK

$$\mathrm{L2RdWrBW}_{Bp(Clk \cdot XeCore)} = \mathrm{L2RdBW}_{Bp(Clk \cdot XeCore)}$$
$$+\mathrm{L2WrBW}_{Bp(Clk \cdot XeCore)}$$

row 79 - L1 READ B EU CLK

$$\mathrm{L1RdBW}_{Bp(Clk \cdot EU)} = \frac{\frac{\mathrm{L1Rd}_B^{(total)}}{|\mathrm{EU}|}}{\mathrm{T}_{clk}^{(total)}}$$

row 77 - L1 READ B XECORE CLK

$$\mathrm{L1RdBW}_{Bp(Clk \cdot XeCore)} = \mathrm{L1RdBW}_{Bp(Clk \cdot EU)} \times |\mathrm{EU}|_{perXeCore}$$

row 80 - L1 WRITE B EU CLK

$$\mathrm{L1WrBW}_{Bp(Clk\cdot EU)} = \frac{\frac{\mathrm{L1Wr}_B^{(total)}}{|\mathrm{EU}|}}{\mathrm{T}_{clk}^{(total)}}$$

row 78 - L1 WRITE B XECORECLK

$$\mathrm{L1WrBW}_{Bp(Clk\cdot XeCore)} = \mathrm{L1WrBW}_{Bp(Clk\cdot EU)} \times |\mathrm{EU}|_{perXeCore}$$

row 83 - L2 READ MAX B XECORE CLK

$$\mathrm{L2RdBW}^{(max)}_{Bp(Clk\cdot XeCore)} = \frac{|\mathrm{XeCore}|_{perXeCU} \times 64}{|\mathrm{XeCore}|_{perXeCU}}$$

row 84 - L2 WRITE MAX B XECORE CLK

$$\mathrm{L2WrBW}^{(max)}_{Bp(Clk\cdot XeCore)} = \mathrm{L2RdBW}^{(max)}_{Bp(Clk\cdot XeCore)}$$

row 85 - L2 READ WRITE MAX B XECORE CLK

$$\mathrm{L2RdWrBW}^{(max)}_{Bp(Clk\cdot XeCore)} = \mathrm{L2WrBW}^{(max)}_{Bp(Clk\cdot XeCore)}$$

row 90 - L2 READ B XECORE CLK PCT

$$\eta_{L2RdBWpXeCore} = \frac{\mathrm{L2RdBW}_{Bp(Clk\cdot XeCore)}}{\mathrm{L2RdBW}^{(max)}_{Bp(Clk\cdot XeCore)}}$$

row 91 - L2 WRITE B XECORE CLK PCT

$$\eta_{L2WrBWpXeCore} = \frac{\mathrm{L2WrBW}_{Bp(Clk\cdot XeCore)}}{\mathrm{L2WrBW}^{(max)}_{Bp(Clk\cdot XeCore)}}$$

row 92 - L2 READ WRITE B XECORE CLK PCT

$$\eta_{L2RdWrBWpXeCore} = \frac{\mathrm{L2RdWrBW}_{Bp(Clk\cdot XeCore)}}{\mathrm{L2RdWrBW}^{(max)}_{Bp(Clk\cdot XeCore)}}$$

row 93 - L1 READ B EU CLK PCT

$$\eta_{L1RdBWpEU} = \frac{\mathrm{L1RdBW}_{Bp(Clk\cdot EU)}}{\mathrm{L1RdBW}^{max}_{Bp(Clk\cdot EU)}}$$

row 94 - L1 WRITE B EU CLK PCT

$$\eta_{L1WrBWpEU} = \frac{\mathrm{L1WrBW}_{Bp(Clk\cdot EU)}}{\mathrm{L1WrBW}^{max}_{Bp(Clk\cdot EU)}}$$

row 106 - L2 HIT RATE ASSUMED RANDOM ACCESS WITHIN THE WORKING DATA SET PCT

$$\mathrm{L2HitRate}_{random\,WS\,access} = \min(\frac{\mathrm{L2Size}_B}{\mathrm{TotalRequiredL2Size}^{1\,instance,\,1\,wave}_{matB\,alwayshit}}, 1)$$

row 107 - L2 MISS RATE PCT

$$\mathrm{L2MissRate} = 1 - \mathrm{L2HitRate}_{random\,WS\,access}$$

row 108 - TOTAL L2 READ TRAFFIC B

$$\mathrm{L2RdB}_{total} = \mathrm{L2Rd}^{(total)}_{B}$$

row 110 - PROBABILITY OF MATA HIT IN L2 DURING A NON FIRST WAVE PCT

$$_{fter\,1st\,wave} = \mathrm{IF}(\left(\mathrm{Size}^{(matA)}_{B} + \mathrm{Size}^{(matB)}_{B} + \mathrm{Size}^{(matC,matD)}_{B}\right) \leq \mathrm{L2Size}_B, 1.0, \mathrm{IF}(\mathrm{K}_{dim} > 2 \times |\mathrm{WorkingSet}|^{(K\,in\,L2)}_{20K\,clks\,of\,thread\,divergen}$$

$$-\frac{\mathrm{K}_{dim} - |\mathrm{WorkingSet}|^{(K\,in\,L2)}_{20K\,clks\,of\,thread\,divergence}}{|\mathrm{WorkingSet}|^{(K\,in\,L2)}_{20K\,clks\,of\,thread\,divergence}}))$$

row 111 - PROBABILITY OF MATB HIT IN L2 DURING A NON FIRST WAVE PCT

$$\mathrm{P}^{(matB)}_{L2Hit,\,after\,1st\,wave} = \mathrm{IF}(\left(\mathrm{Size}^{(matA)}_{B} + \mathrm{Size}^{(matB)}_{B} + \mathrm{Size}^{(matC,matD)}_{B}\right) \leq \mathrm{L2Size}_B, 1.0, 0.0)$$

row 112 - PROBABILITY OF MATA MISS IN L2 DURING A NON FIRST WAVE PCT

$$\mathrm{P}^{(matA)}_{L2Miss,\,after\,1st\,wave} = 1 - \mathrm{P}^{(matA)}_{L2Hit,\,after\,1st\,wave}$$

row 113 - PROBABILITY OF MATB MISS IN L2 DURING A NON FIRST WAVE PCT

$$\mathrm{P}^{(matB)}_{L2Miss,\,after\,1st\,wave} = 1 - \mathrm{P}^{(matB)}_{L2Hit,\,after\,1st\,wave}$$

row 116 - TOTAL HBM READ B AFTER A COMPLETION OF A WAVE CONSIDER COLD CACHE

$$\mathrm{MemRdB}^{(wave)}_{cold\,start} = \frac{\left(\mathrm{MemRdB}^{(wave)}_{cold\,start}\right)_{\mathrm{num}}}{2}$$

$$\begin{aligned}\left(\mathrm{MemRdB}^{(wave)}_{cold\,start}\right)_{\mathrm{num}} = {} & \mathrm{Size}^{(matA)}_{B} \\ & + \mathrm{Size}^{(matA)}_{B} \times \left(\mathrm{CEILING}(\frac{\mathrm{N}_{dim}}{\mathrm{W}^{(XeCUTile)}_{element}}, 1) - 1\right) \times \mathrm{P}^{(matA)}_{L2Miss,\, after\, 1st\, wave} \\ & + \mathrm{Size}^{(matB)}_{B} \\ & + \mathrm{Size}^{(matB)}_{B} \times \left(\mathrm{CEILING}(\frac{\mathrm{M}_{dim}}{\mathrm{H}^{(XeCUTile)}_{element}}, 1) - 1\right) \times \mathrm{P}^{(matB)}_{L2Miss,\, after\, 1st\, wave} \\ & + \left(\left(\mathrm{MemRdB}^{(wave)}_{cold\,start}\right)_{\mathrm{num}}\right)_{\mathrm{aux4}} \times \mathrm{L2MissRate}\end{aligned}$$

$$\begin{aligned}\left(\left(\mathrm{MemRdB}^{(wave)}_{cold\,start}\right)_{\mathrm{num}}\right)_{\mathrm{aux4}} = {} & \mathrm{L2RdB}_{total} \\ & - \left(\left(\left(\mathrm{MemRdB}^{(wave)}_{cold\,start}\right)_{\mathrm{num}}\right)_{\mathrm{aux4}}\right)_{\mathrm{aux1}}\end{aligned}$$

$$\begin{aligned}\left(\left(\left(\mathrm{MemRdB}^{(wave)}_{cold\,start}\right)_{\mathrm{num}}\right)_{\mathrm{aux4}}\right)_{\mathrm{aux1}} = {} & \mathrm{Size}^{(matA)}_{B} \\ & + \mathrm{Size}^{(matA)}_{B} \times \left(\mathrm{CEILING}(\frac{\mathrm{N}_{dim}}{\mathrm{W}^{(XeCUTile)}_{element}}, 1) - 1\right) \times \mathrm{P}^{(matA)}_{L2Miss,\, after\, 1st\, wave} \\ & + \mathrm{Size}^{(matB)}_{B} \\ & + \mathrm{Size}^{(matB)}_{B} \times \left(\mathrm{CEILING}(\frac{\mathrm{M}_{dim}}{\mathrm{H}^{(XeCUTile)}_{element}}, 1) - 1\right) \times \mathrm{P}^{(matB)}_{L2Miss,\, after\, 1st\, wave}\end{aligned}$$

row 117 - TOTAL HBM WRITE B

$$\mathrm{MemWrB}^{(total)} = \mathrm{Size}^{(matC,matD)}_{B}$$

row 118 - TOTAL HBM B

$$\mathrm{MemRdWrB}^{(total)} = \mathrm{MemWrB}^{(total)} + \mathrm{MemRdB}^{(wave)}_{cold\,start}$$

row 121 - HBM READ B CLK

$$\text{MemRdBW}_{BpClk} = \frac{\text{MemRdB}_{cold\,start}^{(wave)}}{\text{T}_{clk}^{(total)}}$$

row 122 - HBM WRITE B CLK

$$\text{MemWrBW}_{BpClk} = \frac{\text{MemWrB}^{(total)}}{\text{T}_{clk}^{(total)}}$$

row 123 - HBM TOTAL B CLK

$$\text{MemRdWrBW}_{BpClk} = \frac{\text{MemRdWrB}^{(total)}}{\text{T}_{clk}^{(total)}}$$

row 126 - HBM BW PCT

$$\eta_{MemBW} = \frac{\text{MemRdWrBW}_{BpClk}}{\text{MemBW}_{BpClk}^{max}}$$

row 127 - GTI READ BW PCT

$$\eta_{GTIRdBW} = \frac{\text{MemRdBW}_{BpClk}}{\text{GtiRdBW}_{BpClk}^{max}}$$

row 128 - GTI WRITE BW PCT

$$\eta_{GTIWrBW} = \frac{\text{MemWrBW}_{BpClk}}{\text{GtiWrBW}_{BpClk}^{max}}$$